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InspireHEP

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Keywords: Standard Model Effective Field Theory, Parton Distribution Functions, Future Colliders, Neural Simulation Based Inference

Professional Experience

since Oct 2024 — **University of Edinburgh, Higgs Centre**, *Postdoctoral Researcher*
- Member of the NNPDF collaboration
- Member of the SMEFiT collaboration

Education

Oct 2020 - Sep 2024 — **VU Amsterdam and Nikhef**, *PhD Theoretical Physics*
- Thesis title: "Fingerprinting New Physics with Effective Field Theories"
  Advisors: Juan Rojo (VU), Wouter Verkerke (UvA)

Sep 2018 - Jul 2020 — **Utrecht University**, *M.Sc. Theoretical Physics*
- GPA 8.0/10.0
- Thesis title "Matching between EFT and UV Complete Models" at Nikhef

Sep 2015 - Jun 2018 — **Utrecht University**, *B.Sc. Physics & Astronomy*
- Graduated cum laude, GPA: 8.8/10.0
- Thesis title: 'Renormalization Group Connected to Neural Networks'

Sep 2017 - Dec 2017 — **University of Edinburgh**, *Erasmus Exchange programme*
- Obtained an average grade of 80 %, 1st Class Honours

Publications

- T. Armadillo, E. Celada, J. ter Hoeve, F. Maltoni, L. Mantani, J. Rojo, A. N. Rossia, S. Tentori, M. O. A. Thomas, and E. Vryonidou, "New Physics Reach through Precision at Future Colliders: a Multi-Pronged Approach," arXiv:2604.16596 [hep-ph].
- R. Barrué *et al.*, "Proton Structure from Neural Simulation-Based Inference at the LHC," arXiv:2604.13157 [hep-ph].
- R. Ball, J. ter Hoeve, and R. Stegeman, "A Determination of the Top Mass from a Global PDF Analysis," arXiv:2603.28865 [hep-ph].
- R. D. Ball, T. Giani, F. Hekhorn, J. ter Hoeve, T. R. Rabemananjara, J. Rojo, R. Stegeman, and M. Ubiali, "Parton distribution functions and theory parameters: an NNPDF perspective," arXiv:2602.24005 [hep-ph].
- J. Altmann *et al.*, *ECFA Higgs, electroweak, and top Factory Study*, vol. 5/2025 of *CERN Yellow Reports: Monographs*. 6, 2025. arXiv:2506.15390 [hep-ex].
- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report: Volume 1, Physics, Experiments, Detectors," *Eur. Phys. J. C* **85** no. 12, (2025) 1468, arXiv:2505.00272 [hep-ex].
- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report: Volume 2, Accelerators, Technical Infrastructure and Safety," *Eur. Phys. J. ST* **234** no. 19, (2025) 5713–6197, arXiv:2505.00274 [physics.acc-ph].

- **FCC** Collaboration, M. Benedikt *et al.*, "Future Circular Collider Feasibility Study Report: Volume 3 Civil Engineering, Implementation and Sustainability," *Eur. Phys. J. ST* **234** no. 17, (2025) 5113–5383, arXiv:2505.00273 [physics.acc-ph]. [Erratum: Eur.Phys.J.ST 234, 33–44 (2025)].
- J. ter Hoeve, L. Mantani, J. Rojo, A. N. Rossia, and E. Vryonidou, "Higgs trilinear coupling in the standard model effective field theory at the high luminosity LHC and the FCC-ee," *Phys. Rev. D* **112** no. 1, (2025) 013008, arXiv:2504.05974 [hep-ph].
- J. ter Hoeve, L. Mantani, J. Rojo, A. N. Rossia, and E. Vryonidou, "Connecting scales: RGE effects in the SMEFT at the LHC and future colliders," *JHEP* **06** (2025) 125, arXiv:2502.20453 [hep-ph].
- J. ter Hoeve, *Fingerprinting New Physics with Effective Field Theories*. PhD thesis, 2025. arXiv:2501.08818 [hep-ph].
- E. Celada, T. Giani, J. ter Hoeve, L. Mantani, J. Rojo, A. N. Rossia, M. O. A. Thomas, and E. Vryonidou, "Mapping the SMEFT at high-energy colliders: from LEP and the (HL-)LHC to the FCC-ee," *JHEP* **09** (2024) 091, arXiv:2404.12809 [hep-ph].
- J. ter Hoeve, E. Laenen, C. Marinissen, L. Vernazza, and G. Wang, "Region analysis of QED massive fermion form factor," *JHEP* **02** (2024) 024, arXiv:2311.16215 [hep-ph].
- J. ter Hoeve, G. Magni, J. Rojo, A. N. Rossia, and E. Vryonidou, "The automation of SMEFT-assisted constraints on UV-complete models," *JHEP* **01** (2024) 179, arXiv:2309.04523 [hep-ph].
- H. La, A. Brokkelkamp, S. van der Lippe, J. ter Hoeve, J. Rojo, and S. Conesa-Boj, "Edge-induced excitations in Bi2Te3 from spatially-resolved electron energy-gain spectroscopy," *Ultramicroscopy* **254** (2023) 113841, arXiv:2305.03752 [cond-mat.mes-hall].
- R. Gomez Ambrosio, J. ter Hoeve, M. Madigan, J. Rojo, and V. Sanz, "Unbinned multivariate observables for global SMEFT analyses from machine learning," *JHEP* **03** (2023) 033, arXiv:2211.02058 [hep-ph].
- A. Brokkelkamp, J. ter Hoeve, I. Postmes, S. E. van Heijst, L. Maduro, A. V. Davydov, S. Krylyuk, J. Rojo, and S. Conesa-Boj*, "Spatially Resolved Band Gap and Dielectric Function in Two-Dimensional Materials from Electron Energy Loss Spectroscopy," *J. Phys. Chem. A* **126** no. 7, (2022) 1255–1262, arXiv:2202.12572 [cond-mat.mtrl-sci].

Seminars and talks

Upcoming	**CMS Top PAG meeting, ATLAS PDF meeting, LHC top WG, COMETA, HEFT2026**
May 2026	**A Determination of the Top Mass from a Global PDF Analysis**, *DIS2026,* (slides)
Dec 2025	**Updates from the SMEFiT collaboration**, *LHC EFT WG* (slides)
Jul 2025	**Towards a combined α_s and m_t determination from a global PDF analysis**, *EPS-HEP* (slides)
Jul 2025	**Connecting scales: RGE effects in the SMEFT at the LHC and future colliders**, *EPS-HEP* (slides)
Jun 2025	**Contributed to E. Laenen's ESPPU symposium opening talk**, *Venice Open Symposium* (slides, nr. 19)
May 2025	**Connecting scales in the SMEFT at the LHC and future colliders**, *Seminar at DAMTP, University of Cambridge* (slides)

Jan 2025	**Single particle extensions at FCC-ee**, *8th FCC Physics Workshop, CERN* (slides)
Dec 2024	**Mapping the SMEFT at future colliders, towards the ESPPU**, *Seminar at IPPP, Durham University*
Jul 2024	**Mapping the SMEFT at High-Energy Colliders: from LEP and the (HL-)LHC to the FCC-ee**, *ICHEP, Prague* (slides)
May 2024	**Update on global SMEFT fits**, *Standard Model at the LHC, Rome* (slides)
Mar 2024	**Global SMEFT fits from (HL)-LHC to future colliders**, *Seminar Radboud University, Nijmegen*
Feb 2024	**Likelihood-free inference for global SMEFT fits**, *1st COMETA General Meeting, Izmir* (slides)
Oct 2023	**Machine Learning opportunities for EFT analyses**, *LHC EFT Working Group, CERN (online)* (slides), (recording)
Oct 2023	**Likelihood learning theory in practice**, *LHC Precision Program, Benasque (Spain)* (slides)
Aug 2023	**Interpreting HEP data in SMEFiT**, *Reinterpretation Forum, Durham* (slides)
Jun 2023	**Statistically optimal observables for global SMEFT fits**, *HEFT, Manchester* (slides)
Dec 2022	**Unbinned multivariate observables for global SMEFT analyses from machine learning**, *CERN, Geneva* (slides)
Sep 2022	**Towards an optimal global SMEFT fit with machine learning**, *Machine Learning at GGI, Florence* (slides)
May 2022	**Unbinned measurements in global SMEFT fits from machine learning**, *Learning to Discover, Paris-Saclay* (slides)
Jan 2022	**Charting Electron Energy Loss Spectra with Machine Learning**, *NWO Physics 2021*
Nov 2021	**Optimally sensitive observables for global EFT fits**, *Dutch Physics Society (NNV) annual meeting* (slides)
Oct 2021	**Optimally sensitive observables for global EFT fits**, *Higgs 2021, Stony Brook University (online)* (slides)

Software packages

- `SMEFiT`, a Standard Model Effective Field Theory Fitter
  `https://smefit.science` (open source, runs on GPUs)
- `NNPDF`, Neural Network Parton Distribution Functions
  `https://docs.nnpdf.science` (open source, runs on GPUs)
- `ML4EFT`, simulation based inference tool based on deep-learning for particle physics data
  `https://lhcfitnikhef.github.io/ML4EFT` (open source)

- `EELSFitter`, an Electron-Energy Loss Spectra analyser
  https://lhcfitnikhef.github.io/EELSfitter (open source)

Referee papers

- **SciPost**: G. Guedes and P. Olgoso, *From the EFT to the UV: the complete SMEFT one-loop dictionary*, 2412.14253.
- **EPJC**: D. Keller, *Experimental Uncertainty Propagation in Neural Network Extraction in Hadronic Physics*, 2509.11456.
- **SciPost**: S. Degenkolb, N. Elmer, T. Modak, M. Mühlleitner and T. Plehn, *A Global View of the EDM Landscape*, 2403.02052.
- **JHEP**: S. Chai, J. Gu and L. Li, *From optimal observables to machine learning: an effective-field-theory analysis of $e^+e^- \to W^+W^-$ at future lepton colliders*, *JHEP* **05** (2024) 292 [2401.02474].

Selected activities and outreach

Since May 2026	**Seminar organiser** Organise the Particle Physics Theory Seminars at the Higgs Centre
Since Feb 2026	**COMETA colloquia organiser** Organise the online COMETA (COmprehensive Multiboson Experiment-Theory Action) colloquia series
Nov 2024	**SMEFT workshop** Organised a SMEFT workshop at the Higgs Centre
Jun 2022 - Sep 2024	**NNV Subatomic Physics council member** Organised scientific events for the Dutch Subatomic Physics Society
Dec 2020 - Sep 2024	**Chair DRSTP PhD council** Organised scientific meetings and social/outreach events for theoretical physics PhD students in The Netherlands
April 2024	**Particle physics outreach talk** Gave a talk for Norwegian high school students, University of Amsterdam
March 2022	**CERN Masterclass** Particle physics outreach workshop for high school students, Nikhef
May 2021	**Guest lecture** Topic: Electron Energy Loss Fitter with Machine Learning, TU Delft

Teaching and supervisory experience

since June 25	**SMEFiT code coordinator** Coordinate code development in the SMEFiT collaboration; lead biweekly coding meetings
since Sep 25	**PhD student supervision** A. Pietraszkiewicz
Sep 22- Jun 23	**Master thesis supervision** P. Herbschleb (2023), ML Optimal Observables in SMEFT **Bachelor thesis supervision** W. Gauthier (2023), J. Bakker and D. Pelan (2021)

since Sep 2020	**Teaching Assistant** Courses, UvA/VU Amsterdam ◦ Effective Field Theories (2023) ◦ Quantum Mechanics II (2021, 2022)
Nov 2016 - Jun 2020	**Teaching Assistant** Courses, Utrecht University ◦ Black Holes (2019, 2020) ◦ Fluid Dynamics (2018, 2020) ◦ Data Analysis (2016, 2017)

Computer and natural languages

- Python, PyTorch/Tensorflow, JAX, C++, Java, Mathematica, Fortran
- Dutch (Native), English (Full professional proficiency)